

AI-Based Anomaly Detection

for Financial Fraud Monitoring

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Problem Statement



Financial Fraud Detection Problem

- Online financial transactions are increasing rapidly.
- Traditional fraud detection systems use predefined rules.
- These systems are weak at detecting new fraud patterns.



Problems caused by fraud

- Financial losses
- Identity theft
- Reputational damage to institutions



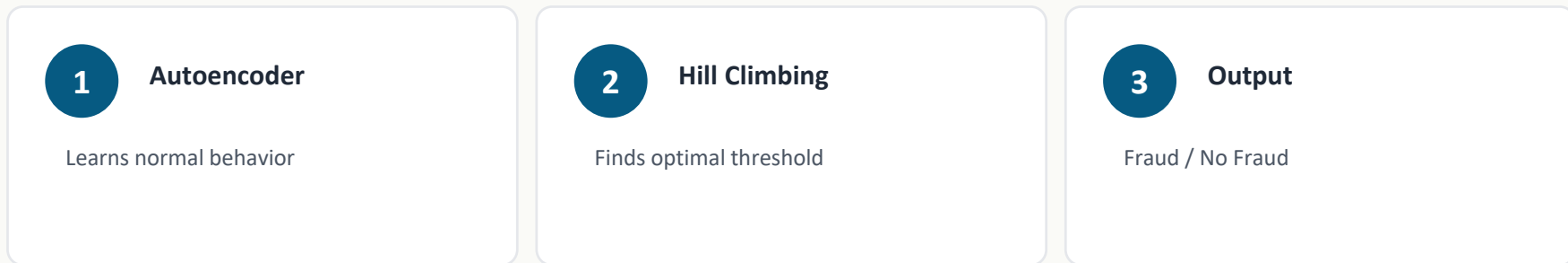
Project Goal

Develop an AI system that detects fraud using:

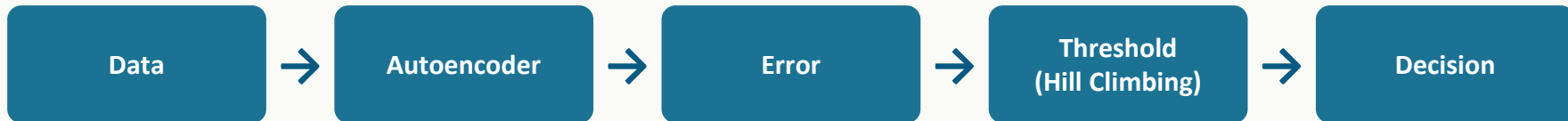
- Anomaly detection (Deep learning)
- Search optimization (Hill climbing)

Our Approach

We built a hybrid AI system

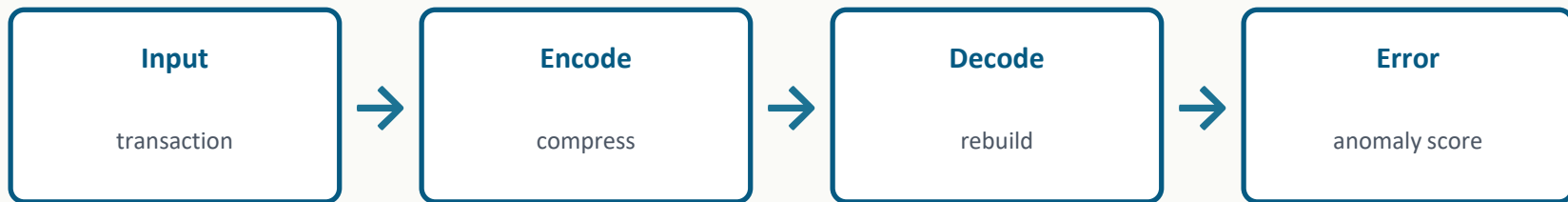


Solution Pipeline



Autoencoder

The autoencoder will learn patterns of normal financial transactions.



Large reconstruction error = unusual behavior

Model Output

- Reconstruction Error = Anomaly Score
- Transactions with higher anomaly scores are more likely to be fraudulent.
- These anomaly scores will be used in the threshold optimization step.
- Learns what “normal” transactions look like.

Hill Climbing Implementation

1

Local Search Algorithm

Look at all the neighbors and move to a state that maximizes/minimizes an evaluation function (F1-score). Does not guarantee a global optimum.

2

Goal

Find the optimal classification threshold value to decide FRAUD or NO FRAUD.

3

Metric to Improve

F1 score → Minimize False Negatives!

Hill Climbing — Implementation & Challenges



Setup

State = threshold value

Neighbors = small changes in threshold

Goal = maximize F1-score



Implementation Challenges

a. Need to choose a better utility function

b. Locally optimal value

Decision Pipeline

Autoencoder Reconstruction Error



Threshold Score (Fixed)



Fraud / No Fraud

Training Results

Model Performance Comparison (Baseline vs Optimized)

Fig: F1 score as the Utility Function

Metric	Baseline	Optimized
Accuracy	0.96	0.99
Precision	0.14	0.55
Recall	0.84	0.79
F1 Score	0.25	0.65

Fig: Recall as the Utility Function

Metric	Baseline	Optimized
Accuracy	0.96	0.86
Precision	0.14	0.05
Recall	0.84	0.88
F1 Score	0.25	0.10

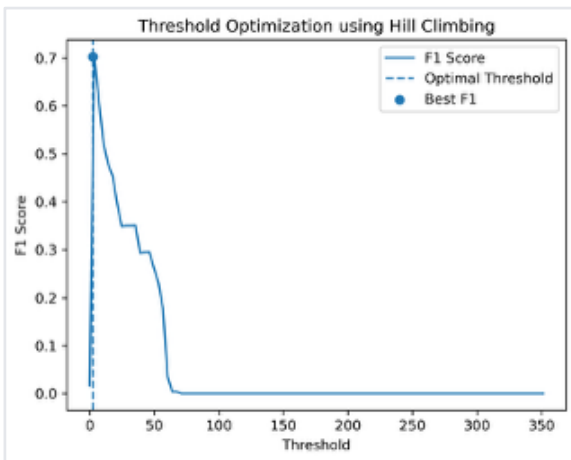
- Optimization improves precision significantly
- Recall slightly decreases but remains high
- F1-score improves → better balance

Threshold Optimization & Precision–Recall



F1 Score vs Threshold (Hill Climbing)

F1 Score



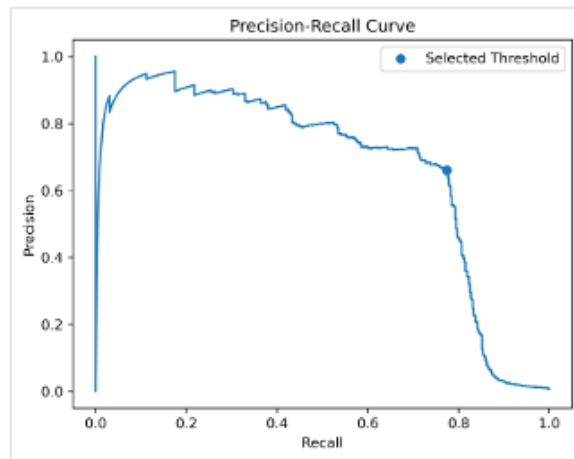
Threshold →

- Optimal threshold identified using Hill Climbing to maximize F1-score
- F1-score peaks at a specific threshold, then drops sharply.



Precision–Recall Curve

Precision



Recall →

- PR curve highlights trade-off between detecting fraud (recall) and reducing false alarms (precision)
- Strong recall at lower thresholds, but precision decreases as recall increases

DEMO

Live walkthrough

Timeline

Weeks 1–2

Data Preparation

Weeks 3–4

Model Implementation

Weeks 5–6

**Model Analysis, Result
Visualization and Fine
Tuning**

Weeks 7–8

**Fine Tuning, Demo
Preparation**

What Worked Well & Challenges

What Worked Well



Hybrid Approach

Classical AI Algo: Decision Making
Deep Learning: Feature Extraction



Beating the Imbalance Problem



Metric Improvement

Challenges



The Precision-Recall Tug-of-War



Search Algorithm: Missing Global Optima



Sophisticated Utility Function



Security

What We Learned

- 1 **Correct Model Formulation**
- 2 **Accuracy is a Trap**
- 3 **Choice a Problem Specific Metric**
- 4 **Hybrid Approaches Wins**
- 5 **Tradeoff between different Search Algorithms**

Ethics and Reflections

The “Black Box” Problem



The Cost of False Positives

Institutional Reputation



Need for Responsible Deployment

Adversarial Vulnerability: Data Poisoning, GANs.



Hidden Bias in “Normal” Data

Learns what is “normal” from historical data, it may inherently penalize edge-case demographics.



Data Privacy and Over-Surveillance

Addressing Feedback Questions



What limitations may the system have? Is it possible for a criminal to circumvent the system, and if so, how can you address this limitation?



Dynamic Thresholding

- Frequently Rerun Search Algorithm: Keep the threshold up to date
- Continuous Learning Pipelines



Ensemble Approach

Autoencoder + supervised classifier to catch known fraud signatures.

Contributions

Asim: Problem Modeling, Search Algorithm, Data Visualization, Fine Tuning

Zain: Autoencoder Implementation, Proposed Modelling, Report Making, Fine Tuning

Khwabena: Coding, Implementation, Demo, Slides Hill Climbing

Any Questions?

Thank you!